

ABM BUSINESS LOGIC BIBLE

§6 v3 (INFORMATION ECONOMICS) — KNOWLEDGE-ACQUISITION ENGINE

Iterations 4–13 · 10 Architectural Layers · 28 New Rules · Phase Flags · 1 New OPEN-Q

§6 v1 specified the 7-step workflow. §6 v2 unmasked three semantic illusions (reachability, initiatives, staleness).

§6 v3 reframes the engine itself. §6 is no longer an enrichment workflow — it is a **knowledge-acquisition engine operating under cost, time, and uncertainty constraints**. Its purpose is not maximum data collection; it is **maximum uncertainty reduction per unit of investment**. Ten architectural layers govern: what to learn, what not to learn, when learning should stop, how knowledge compounds across accounts, how researcher biases distort reality, and which uncertainties are fundamentally unknowable.

Iterations	10 (4–13)	Architectural layers	10
New rules	28	Phase 1 / 2 split	≈12 / 16
Reframe	Workflow → Knowledge-Acquisition Engine	OPEN-Q added	6.5 (budget unit)

THE FRAMING SHIFT — §6 IS NOW AN ECONOMIC SYSTEM, NOT A WORKFLOW

Up to §6 v2, enrichment was treated as a workflow that produces output. §6 v3 reframes it as the engine's **information-acquisition economy**: every potential enrichment action is an investment decision; every missing fact carries a marginal value; some uncertainties are irreducible and must be marked as such rather than chased indefinitely. The architectural test from now on: every §6 rule must be interpretable as either an information-acquisition cost, an information-value estimate, or a stopping/governance constraint on the acquisition economy.

THE TEN LAYERS — §6 v3 ARCHITECTURE MAP

#	Layer	From iter.	What it adds
1	Marginal Information Value (Comp)	4	Diminishing-returns stopping; perfect dossiers are economically irrational
2	Blocking vs Non-Blocking (Block)	5	Missingness has structure; prioritize what blocks decisions
3	Search Stopping (Stop)	6	Enrichment exists for decision support, not knowledge accumulation
4	Information Quality vs Quantity (Qual)	7	One high-quality fact beats fifty low-quality; track Decision Utility
5	Multi-Layer Decay (Decay)	8	Operational / Tactical / Strategic / Structural — four time horizons
6	Information Economics (Econ)	9	Information value = function of decision value; per-account intelligence budget
7	Search Strategy (Search)	10	Information has dependency chains; Keystone Unknowns unlock many downstream
8	Knowledge Portfolio (Port)	11	Knowledge has network effects; Shared Intelligence Assets; exclusivity respected
9	Researcher Bias (Human)	12	Availability / Confirmation / Completion / Freshness counter-disciplines
10	Reducible vs Irreducible (Budget)	13	Residual Uncertainty Register; the engine stays honest about what it cannot know

KNOWLEDGE PORTFOLIO AND VENDOR EXCLUSIVITY — NOT A CONTRADICTION

Layer 8 says ecosystem intelligence (Mastercard, KPMG, regulatory advisory) researched for SNB is reusable across Riyadh, Al Rajhi, Alinma. The Bible's vendor-exclusivity discipline (Backbase = SNB only; Efidence = Riyadh only) says per-account vendor facts must NOT propagate. These are different categories: **connector and ecosystem intelligence is reusable; per-account vendor relationships are exclusivity-bound**. The two rules co-exist by construction: Shared Intelligence Assets are explicitly scoped to ecosystem facts, never to bank-specific vendor relationships.

LAYER 1 — MARGINAL INFORMATION VALUE (COMP)

The first contact found is worth a lot. The 10th is worth very little. §6 v1 had no concept of diminishing returns; this layer makes it explicit and turns it into a stopping rule.

Rule ID	Phase	Specification
ENR-COMP-01	P1	Every potential enrichment action carries an Expected Information Gain estimate computed BEFORE execution. Finding a missing committee contact when the dossier has 2 known contacts has high estimated gain; finding the 8th has low. The estimate is the gate.
ENR-COMP-02	P1	Enrichment STOPS when Expected Information Gain < Acquisition Cost — not when every dossier field is filled. Perfect dossiers are economically irrational. The 7-step workflow runs to completion only when each remaining step still clears the marginal-value gate.
ENR-COMP-03	P2	Marginal-value estimates are calibrated against actual decision outcomes (Tier 6 counterfactual): if a step rated 'high gain' produced no decision change, its marginal-value model is adjusted.

LAYER 2 — BLOCKING vs NON-BLOCKING (BLOCK)

Rule ID	Phase	Specification
ENR-BLOCK-01	P1	Every dossier field is classified at design time as Blocking or Non-Blocking . Blocking: buying committee unknown, primary decision-maker unknown, no outreach path, no verified contact. Non-Blocking: secondary phone, office location, biography details.
ENR-BLOCK-02	P1	Enrichment prioritization is Blocking-Information-First , not completeness percentage. The engine fills blocking gaps before any non-blocking, regardless of how easy a non-blocking field would be (closes the availability-bias trap).
ENR-BLOCK-03	P1	Blocking-field gaps trigger LOW_COMMITTEE_COVERAGE and equivalent per-field flags; the flag binds §4.3 Reachability scoring and gates EXEC stage 6. Non-Blocking gaps do not gate downstream stages.

LAYER 3 — SEARCH STOPPING (STOP)

Without an explicit stopping condition, enrichment becomes endless research — the failure mode every human research team eventually hits.

Rule ID	Phase	Specification
ENR-STOP-01	P1	Every enrichment workflow carries an explicit Stopping Condition . Default for HOT: buying-committee coverage $\geq 80\%$ AND ≥ 2 reachable contacts (Response Reachability per §6 v2) AND ≥ 1 outreach path resolved. When all three pass, enrichment stops.
ENR-STOP-02	P1	Additional enrichment beyond the stopping threshold requires explicit justification: a named manual trigger or an event-driven trigger (§6 v2 ENR-REF-DEEP-01). Continuing past stopping by default is forbidden.
ENR-STOP-03	P1	Stopping is observable: every enrichment cycle logs whether it stopped on the threshold (clean), on marginal-value gating (Layer 1), on cost ceiling (Layer 6), or on irreducibility (Layer 10). Stop-reason is a first-class telemetry field.

LAYER 4 — INFORMATION QUALITY vs QUANTITY (QUAL)

Rule ID	Phase	Specification
ENR-QUAL-01	P2	Every enriched fact carries an Information Quality Score : decision relevance, uniqueness, actionability, confidence (RCM stamp), recency. The Quality Score is stored alongside the fact and consumed by the Layer 6 ROI engine.
ENR-QUAL-02	P2	Enrichment success is measured by Decision Utility — how much the acquired information changed downstream decisions — not by Record Volume. KPI 'records enriched per account' is replaced by 'decision-impact-weighted information acquired'.
ENR-QUAL-03	P1	A smaller, higher-quality dossier is preferred to a larger, lower-quality one. The 7-step workflow is configurable: a step that consistently produces low-Quality output for a segment may be skipped without completeness penalty.

LAYER 5 — MULTI-LAYER DECAY (DECAY)

Layer	Category	Typical decay	Examples
Operational	Contact-level facts	7–30 days	Email validity, phone, current title
Tactical	Activity-level facts	30–90 days	Active hiring, recent press, current vendor activity
Strategic	Priority-level facts	6–12 months	Strategic priorities, transformation initiatives, regulatory posture
Structural	Identity-level facts	3–5 years	Organizational DNA, founding story, core market positioning

Rule ID	Phase	Specification
ENR-DECAY-01	P1	Every fact is tagged with one of four decay layers: Operational (7–30d), Tactical (30–90d), Strategic (6–12m), Structural (3–5y). Tag is set at acquisition; categories are not negotiable per fact.
ENR-DECAY-02	P1	Decay state is read at fact-consumption time (EPIS-HALF-02), per category. Strategic and Structural facts remain authoritative far longer than the V2 30-day cadence implies — §6 v2 ENR-REF-DEEP-04 turns 30d into a floor.
ENR-DECAY-03	P1	§6 v2 Iteration 3 Ramadan/Eid 1.5× decay-rate adjustment applies to Operational and Tactical, NOT Structural — organizational identity does not decay faster during Ramadan.

LAYER 6 — INFORMATION ECONOMICS (ECON)

Cost-tiered enrichment caps spend by tier. Layer 6 goes deeper: it ties spend to the **expected value of the decision the information supports**. A \$5M flagship-bank opportunity deserves a larger intelligence budget than a \$50K commercial account, even if both are HOT.

Rule ID	Phase	Specification
ENR-ECON-01	P2	Every account receives a Maximum Intelligence Budget = $f(\text{Expected Opportunity Value} \times \text{Stage Probability} \times \text{Strategic Importance})$. A \$5M strategic bank receives a larger budget; the budget is dynamic and updates with the underlying inputs.
ENR-ECON-02	P2	Enrichment spend is tracked as Intelligence ROI = $\text{Decision Improvement} \div \text{Information Cost}$. This KPI replaces 'API cost per account' as the engine's optimization target.
ENR-ECON-03	P2	Research budget dynamically INCREASES for: strategic accounts, high-uncertainty accounts (low EPIS coverage), and high-value opportunities. Uniform per-tier caps are Phase 1 only; Phase 2 routes by per-account economics.

OPEN-DESIGN-QUESTION OPEN-Q6.5 — BUDGET UNIT OF ACCOUNT

Question: Is the Maximum Intelligence Budget (ENR-ECON-01) denominated in *money* (real currency) or in *attention units* (the \$5 v3 TIER-ATTN-001 currency)?

Candidate options: (a) **Money** — each account has a real-dollar budget; aligns with finance reporting. (b) **Attention units** — budget priced in the same unit-of-account as \$5 v3; cleaner architectural unification. (c) **Both** — dual ledgers: money for finance, attention for engine-internal allocation.

Trade-off: (a) is what finance wants. (b) is what the architecture wants. (c) is what reality probably needs once the engine matures.

Dependent rules: ENR-ECON-01..03. Engineering safe default: ship attention-units (b) at Phase 1 for architectural unification; expose money-conversion view in OBS for finance reporting; revisit (c) at 90-day review.

LAYER 7 — SEARCH STRATEGY (SEARCH)

Most enrichment systems search in workflow order. The engine should search in **dependency order**: finding the initiative owner often reveals the buying committee, which reveals the path, which reveals the outreach target. One well-chosen search unlocks four answers.

Rule ID	Phase	Specification
ENR-SEARCH-01	P2	Every enrichment target carries a Dependency Value — the count of downstream unknowns this fact could resolve. A fact with Dependency Value = 4 is more valuable than a fact with Dependency Value = 0.
ENR-SEARCH-02	P2	Search order is optimized by Expected Information Gain × Dependency Value ÷ Acquisition Cost . The engine selects the next search that maximizes this ratio, not the next step in the workflow's literal order.
ENR-SEARCH-03	P2	The engine identifies and tracks Keystone Unknowns — missing facts whose resolution unlocks multiple downstream decisions. The Primary Initiative Sponsor is typically a Keystone Unknown for a HOT bank account.

LAYER 8 — KNOWLEDGE PORTFOLIO (PORT)

Rule ID	Phase	Specification
ENR-PORT-01	P2	Every intelligence artifact carries a Reusability Score . Ecosystem facts (connector intelligence, regulatory landscape, advisory networks) score high; per-account vendor relationships score zero (exclusivity-bound).
ENR-PORT-02	P2	Knowledge acquisition priority increases when Expected Cross-Account Reuse is high. Researching Mastercard KSA leadership (high reuse) is prioritized over a single local manager at one account (low reuse) at equivalent immediate value.
ENR-PORT-03	P2	Shared Intelligence Assets are first-class objects, separate from per-account dossiers: KSA banking ecosystem map, Open Banking vendor map, Regulatory intelligence map, Advisory council graph. Queried as supplementary context, not duplicated into each dossier.
ENR-PORT-04	P1	Vendor exclusivity discipline binds: per-account vendor facts (Backbase=SNB only, Efigence=Riyad only) are NEVER promoted into Shared Intelligence Assets. The Reusability classifier hard-rejects any fact tagged as bank-specific-vendor.

LAYER 9 — RESEARCHER BIAS DISCIPLINE (HUMAN)

These rules govern the human-in-the-loop research process and the AI agents that emulate it. Researchers carry systematic biases that destroy intelligence quality: availability bias (research what's easy), confirmation bias (find supporting evidence), completion bias (fill fields rather than resolve uncertainty), freshness bias (overvalue recent info).

Rule ID	Phase	Specification
ENR-HUMAN-01	P1	Research tasks (human and AI) must explicitly ask: what would disprove the current working hypothesis? The disconfirmation question is a required field in any structured research output for HOT accounts. Counters confirmation bias.
ENR-HUMAN-02	P1	Research completion is measured by uncertainty reduction, not by field completion. A dossier with every field filled but no critical-unknown resolved is incomplete; a dossier with three fields filled that resolved all blocking unknowns is complete.
ENR-HUMAN-03	P1	For major accounts (strategic or flagship tier), the dossier is not marked complete until a Contradictory Evidence Check has been performed and logged: 'we looked for evidence against the working narrative; here is what we found / didn't find.'
ENR-HUMAN-04	P1	Availability bias counter: workflow order is set by Layer 7 (Dependency Value), not by what's easiest to find. The engine MAY override this order only with a logged justification.

LAYER 10 — REDUCIBLE vs IRREDUCIBLE UNCERTAINTY (BUDGET)

Rule ID	Phase	Specification
ENR-BUDGET-01	P1	Every identified unknown is classified at design time: Reducible (more research will likely resolve), Difficult (resolvable but expensive), or Currently Unknowable (no public source path; do not chase). Classification is a first-class field, not a comment.
ENR-BUDGET-02	P1	Research stops on an unknown when Expected Cost of Further Reduction > Expected Value of the reduced uncertainty. Unknowable items stop immediately; Difficult items stop at their cost ceiling; Reducible items continue until the Layer 1 marginal-value gate binds.
ENR-BUDGET-03	P1	Every account dossier maintains a Residual Uncertainty Register: the named list of things the engine has decided it cannot currently know (e.g. 'budget status unknown', 'executive private preference unknown', 'internal resistance unknown'). The Register is the operational counterpart to EPIS-COV-03 and binds downstream: §7 personalization cannot ground a message in a fact present in the Register.

LAYER 10 IS WHY THE ENGINE STAYS HONEST

Without an explicit Residual Uncertainty Register, the engine eventually starts to *infer* the things it can't know — hallucinating budget posture, guessing at executive preference, manufacturing political-resistance narratives. Layer 10 forces those gaps to be permanent, named, and downstream-binding. The operational twin of EPIS-RCM-05 (anti-fabrication) applied to the dossier itself.

EXHAUSTION ASSESSMENT — §6 AFTER THREE PASSES

Area	§6 v1	§6 v2	§6 v3	Frontier remaining
Workflow logic	~95%	~95%	~98%	—
Reachability theory	~50%	~80%	~95%	—
Initiative analysis	~70%	~80%	~95%	—
Refresh architecture	~85%	~75%	~95%	Resolve OPEN-Q6.4 empirically
Information economics	—	—	~90%	Resolve OPEN-Q6.5 (budget unit)
Search strategy	—	—	~90%	Keystone Unknown calibration with outcome data
Knowledge reuse	—	—	~90%	Shared Intelligence Asset governance + access control
Researcher bias	—	—	~85%	Bias detection telemetry; AI-researcher bias models
Reducible / irreducible	—	—	~90%	Classifier calibration over time

STOPPING ON §6 — RECOMMENDED

§6 v3 lands the architecture at ~95% completeness. The next genuine class of discoveries lives in §7 — where all the intelligence acquired by §6 must be converted into decisions, narratives, and persuasion, and where entirely new failure modes appear (message-strategy hallucination, narrative drift, persona-mismatch, cross-channel inconsistency). §6 is recommended-stopped at v3.

§6 v3 RULE ROLL-UP & PLACEMENT STATUS

Layer	Rules	Phase	Cross-section binding
1 Comp	ENR-COMP-01..03	P1 / P2	Scarcity Economics, Tier 6 counterfactual
2 Block	ENR-BLOCK-01..03	P1	§4.3 Reachability, EXEC stage 6 gating
3 Stop	ENR-STOP-01..03	P1	SIG-PATH P1–P5, §6 v2 Response Reachability, OBS
4 Qual	ENR-QUAL-01..03	P2 / P1	EPIS-RCM stamp, Tier 6 Decision Utility
5 Decay	ENR-DECAY-01..03	P1	EPIS-HALF, §6 v2 ENR-REF-DEEP-04
6 Econ	ENR-ECON-01..03	P2	§5 v3 attention budget (OPEN-Q6.5)
7 Search	ENR-SEARCH-01..03	P2	Dependency graph, Keystone Unknown identification
8 Port	ENR-PORT-01..04	P2 / P1	Vendor exclusivity (SIG-VENDOR-01)
9 Human	ENR-HUMAN-01..04	P1	M0 hallucination governance, §7 Agent prompts
10 Budget	ENR-BUDGET-01..03	P1	EPIS-COV-03, EPIS-RCM-05, §7 grounding gate

§6 v3 COMPLETE — WHAT IT ESTABLISHED

§6 is no longer an enrichment workflow. It is the engine's **Knowledge-Acquisition Engine**: an economic system that decides what is worth knowing, what blocks decisions, when to stop searching, how to value quality over quantity, how knowledge decays across four time horizons, how spend is tied to opportunity economics, how to search in dependency order, how knowledge compounds across the portfolio, how researcher biases are countered, and how reducible uncertainty is separated from irreducible. Ten layers, 28 new rules.

The architectural test: every §6 rule must be interpretable as an information-acquisition cost, value estimate, or stopping constraint.

PLACEMENT MAP UPDATE

§6 status: **Complete-with-Opens** (OPEN-Q6.1–6.5 — four prior plus the new 6.5 on budget unit). §6 is recommended-stopped at v3 per the exhaustion assessment; further depth would more productively go into §7.

ON THE HORIZON

§7 AI Personalization Engine — the 7-agent architecture, temperatures, language rules, hard message constraints — bridged to M0 hallucination governance, EPIS RCM grounding, API-09, §6 v2's two-layer initiative input, and §6 v3's Residual Uncertainty Register (Layer 10) as a grounding gate.